APRIOPRI ALGORITHM

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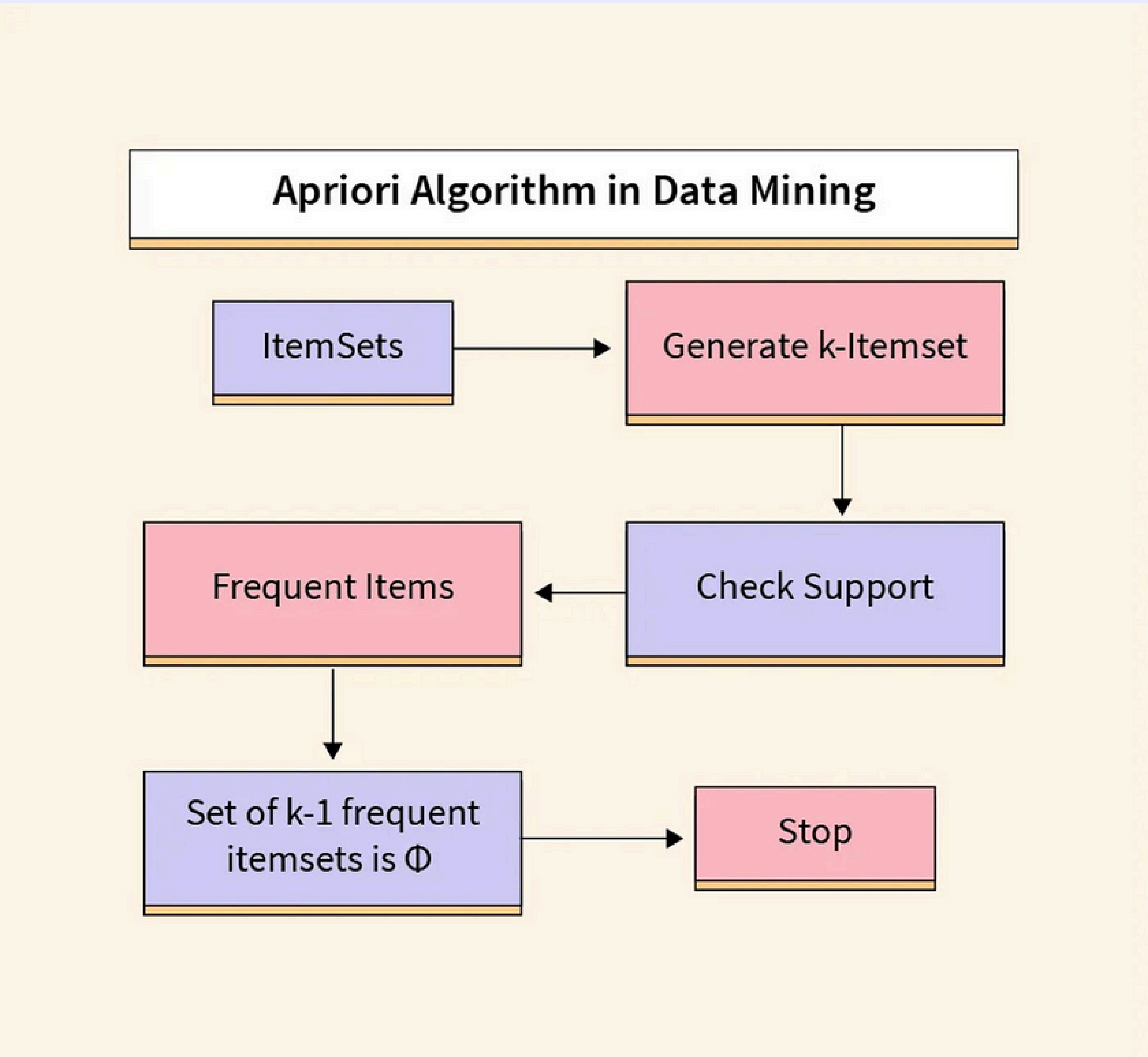
OVERVIEW

Apriori is a classic algorithm to find frequent itemsets and generate association rules. It uses the principle that all subsets of a frequent itemset must also be frequent. Key metrics include:
- **Support**: Frequency of an itemset
- **Confidence**: Likelihood of rule occurrence
- **Lift**: Strength of association

GENERATING CANDIDATE ITEMSETS

1. **Transform Data**: Convert each loan record into a set of attribute-value pairs. (e.g., {Gender=Male, Married=Yes, Education=Graduate})
2. **Generate to C1**: Count support of individual items.
3. **Prune to L1**: Remove items below support threshold.
4. **Generate C2**: From pairs from L1, count their support.
5. **Repeat**: Continue for C3, C4,... until no frequent itemsets remain.

Figure 1: Understanding the Apriori Algorithm in Data Mining



PRUNE NON-FREQUENT ITEMSETS

Apply the Apriori algorithm to:
- Identify frequent itemsets
- Combined itemsets like {'loan_status=0', 'previous_loan_defaults_on_file=Yes'} show frequent co-occurrence of values.
- Remove or prune combinations that don't meet the minimum support threshold

GENERATE ASSOCIATION RULES

- Perform preprocessing and generating frequent itemsets on a selected subset of features, like loan_status and previous_loan_defaults.
- From the frequent itemsets, generate rules like: IF person_gender=male AND loan_intent= EDUCATION THEN loan_status=1
- Use metrics like confidence and lift

RESOURCE PAGE

ADVANATGE OF APRIORI ALGORITHM

- Simplicity & ease of implementation
- Works well on unlabelled data
- Uses simple pruning to improve speed
- Flexible (supports multiple thresholds)
- Extensions for various use cases can be created easily
- Interpretable and easy to understand

DISADVANATGE OF APRIORI ALGORITHM

- Computational complexity
- Time & space overhead
- Difficulty handling sparse data
- Limited discovery of complex patterns
- Higher memory usage
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- Inability to handle numeric data
- Lack of incorporation of context

APPLICATIONS OF ASSOCIATION RULES

- Market Basket Analysis: e.g., "If people buy bread, they often buy butter."
- Web Usage Mining: Predict next pages a user might visit.
- Healthcare: Identify co-occurrence of symptoms/diseases.
- Fraud Detection: Detect unusual associations in transaction patterns.
- Recommendation Systems: Suggest products based on frequent combinations.

CHALLENGES IN ASSOCIATION RULES MINING

- Scalability: Huge datasets → computationally expensive.
- Rare Item Problem: Rare but important itemsets may be pruned due to low support.
- Redundant Rules: Too many rules generated, many redundant or obvious.
- Interestingness: Not all frequent rules are meaningful; requires evaluation metrics.
- Dynamic Data: Frequent rescanning needed when data changes.
- Handling Quantitative Data: Harder when data isn't categorical (e.g., age, price).

$$Rule:\ X \Rightarrow Y$$

$$Support = \frac{frq(X,Y)}{N}$$

$$Confidence = \frac{frq(X,Y)}{frq(X)}$$

$$Lift = \frac{Support}{Supp(X) \times Supp(Y)}$$